

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Business Administration (MBA)

Semester – II University Examination May 2010

MB 709: QUANTITATIVE ANALYSIS FOR MANAGEMENT- 2 (QUAM – 2)

Date: 12 May 2010, Wednesday · Time: 12.30 PM to 03.30 PM Maximum Marks: 70

Instructions:

1. This paper contains six questions; the last one is a case analysis.
2. Answer all questions.
3. The first five questions carry 10 marks each and the case analysis carries 20 marks.

- Q1. A company has three production units at X, Y and Z manufacturing a make-to-stock product which the units send to its two warehouses W_1 and W_2 . The warehouses, in turn, regulate the supply of the product to the four markets denoted by A, B, C and D.

The following table (Table 1) presents the data on the capacities of the manufacturing units and the cost of manufacture and transportation of the product from the manufacturing units to the warehouses:

Table 1: Manufacturing Units, their capacities and Cost of Production and Transportation

Manufacturing Unit	Cost of manufacture and Transportation to Warehouse (Rs. per unit)		Capacity (Number of units in '000)
	W_1	W_2	
X	20	10	150
Y	15	10	100
Z	10	12	100
	Total		350

Table 2 below shows the demand at the four markets and the cost of transportation per unit from each of the warehouses to the four markets:

Table 2: Market Demand and Cost of Transportation from Warehouses to Markets

Warehouse	Cost of Transportation per unit from Warehouse to Markets (Rs. per unit)			
	A	B	C	D
W_1	8	9	7	14
W_2	8	7	11	12
Demand (Number of units in '000)	80	100	70	100

Formulate a Linear Programming Problem for minimizing the total cost of production and distribution. [10]

- Q2. The Board of Directors of General Wheels Co. is considering seven large capital investments. Each investment can be made only once. These investments differ in the estimated long-run profit (net present value) that they will generate as well as the amount of capital required, as shown by the following table (in units of millions of rupees):

	Investment Opportunity						
	1	2	3	4	5	6	7
Estimated long-run profit (NPV)	170	100	150	190	70	130	90
Capital required	430	280	340	480	170	320	230

The total amount of capital available for these investments is Rs.1,000 million. Investment opportunities 1 and 2 are mutually exclusive, and so are 3 and 4. Furthermore, neither 3 nor 4 can be undertaken unless investment 1 or 2 is undertaken. There are no such restrictions on investment opportunities 5, 6 and 7. The objective is to select the combination of capital investments that will maximize the total estimated long-run profit (NPV).

- (a) Formulate a binary integer linear programming model for this problem. [5]
- (b) The following goals are set for the above investment decision:

Goal 1: The investment should be *as close to* Rs. 1,000 million *as possible*.

Goal 2: A *minimum* NPV of Rs. 450 million is to be achieved.

Goal 3: A portfolio of three or more investments is *desirable*.

The above goals have to be achieved in the order given above.
Set up a Goal Programming model for investment decision.

[5]

- Q3. There are ten possible roads that a truck can traverse to reach destination '8' from origin '1', as shown in the following table:

Branch	Start Node	End Node	Distance (in hundreds of km.)
Branch 1	1	2	3
Branch 2	1	3	2
Branch 3	2	4	3
Branch 4	3	5	3
Branch 5	4	5	1
Branch 6	4	6	4
Branch 7	5	7	2
Branch 8	6	7	2
Branch 9	6	8	3
Branch 10	7	8	6

Each road can be considered a branch in the shortest route problem. Draw the network diagram and determine the best way to get from origin '1' to destination '8' that will

minimise the total distance travelled. All distances are in hundreds of km. [10]

Q4. Patients arrive at a dentist's clinic at a mean rate of 2.8 patients per hour. The dentist can treat patients at a mean rate of 3 patients per hour. A study of patient waiting time shows that, on an average, a patient waits 30 minutes before seeing the dentist.

(a) What is the average number of patients in the waiting room? [4]

(b) If a patient arrives at 10.10 AM, when can he be expected to leave the clinic? [6]

Q5. A company planning to sell gift packets during Deepavali season proposes to combine dry fruits (A) and nuts (B). Each gift packet must weigh 5 kg. and should contain at least 2 kg. of A and no more than 4 kg. of B. The net profit contributions to the company from the two ingredients A and B are Rs. 25 and Rs. 30 per kg. respectively. The MSExcel formulation and solution of the problem are given in the following tables:

	A	B			
	25.00	30.00			
Ingredients	1.00	1.00	5.00	=	5.00
IngA	1.00	0.00	2.00	>=	2.00
IngB	0.00	1.00	3.00	<=	4.00
	2.00	3.00			140.00

Microsoft Excel 11.0 Answer Report

Worksheet:

[Book1]Sheet1

Report Created: 5/11/2010 6:39:37 AM

Target Cell (Max)

Cell	Name	Original Value	Final Value
\$H\$14	TotalProfit	140.00	140.00

Adjustable Cells

Cell	Name	Original Value	Final Value
\$D\$14	A	2.00	2.00
\$E\$14	B	3.00	3.00

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$F\$8	Ingredients	5.00	\$F\$8=\$H\$8	Not Binding	0
\$F\$9	IngA	2.00	\$F\$9>=\$H\$9	Binding	0.00
\$F\$10	IngB	3.00	\$F\$10<=\$H\$10	Not Binding	1

Microsoft Excel 11.0 Sensitivity Report

Worksheet: [Book1]Sheet1

Report Created: 5/11/2010 6:39:37 AM

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$D\$14	A	2.00	0.00	25	5	1E+30
\$E\$14	B	3.00	0.00	30	1E+30	5

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$F\$8	Ingredients	5.00	30.00	5	1	3
\$F\$9	IngA	2.00	-5.00	2	3	1
\$F\$10	IngB	3.00	0.00	4	1E+30	1

You are required to:

- (a) Formulate the problem algebraically. [2]
- (b) Solve it graphically. [4]
- (c) Explain the shadow prices of Ingredients, IngA and IngB. [4]

Q6. Raj and Adit were classmates in an engineering programme during the years 2002-06. Adit took up a job with a mechanical engineering firm dealing with the manufacture, assembly, commissioning and maintenance of air-conditioners and refrigeration equipment. Raj, on the other hand, decided to pursue his higher education with a leading B-school and graduated as an MBA. Fired by imagination, creativity and innovation and driven by a spirit of entrepreneurship, Raj decided to venture on his own, instead of seeking placement through his institution. He remembered Adit's expertise in engineering and met him with his proposal to go in for a small-scale industry that could be a sub-contractor to the large firm in which Adit was employed.

It did not take long for both of them to join hands and set up a partnership (RAJDIT) for manufacture of brass and copper bushes and sell them to the manufacturers of air-conditioning and refrigeration systems. The business has been nearing a year of completion, when Raj and Adit decided to take a break from their factory and business setting and go for a retreat for calm reflection of their business experience in the last one year or so.

Like any start-up business, RAJDIT had its cup of joy and woes. There were good times where the sale was good, inventory was low, supplies were good, cash inflow and outflow were balanced and net cash flow was positive, debtors were responsive, creditors were sympathetic and the mood was upbeat. There were also moments of depression, marked by poor sales, severe cash crunch, mounting debtors, pressing creditors, huge inventory, etc. While reflecting on whether the business is likely to lead to long-term profitability, they looked at some data that Raj had painfully collected and collated in the form of some probability distributions.

The following table gives the probability distributions developed by Raj:

Selling Price		Unit Production Cost		Sales Volume per week		Fixed Expenses	
Selling Price (Rs./unit)	Probability	Unit Cost (Rs.)	Probability	Sales Volume (Units)	Probability	Fixed Expenses (Rs.)	Probability
5.00	0.20	2.50	0.35	15,000	0.30	20,000	0.50
5.50	0.50	3.00	0.50	18,000	0.45	25,000	0.30
6.00	0.30	3.50	0.15	20,000	0.25	30,000	0.20

Raj and Adit, sipping their coffee in between their sandwich bites, looked at the table by turns and were wondering whether they had chosen a profitable product line or they should think of a fresh business strategy. Raj was recollecting several topics from his MBA such as break-even analysis, cash flow analysis, 4Ps, decision analysis, LPP, Goal Programming, ILPP, networks and network flows, simulation, etc. He started sharing his lessons with his partner hoping to have some new insights.

Raj was showing Adit a table of random numbers which is included at the end of this question.

How can you help RAJDIT?

[20]

100 Random Numbers*

57966 86401 01096 03896 22054 67579 57561 14172 17781 80397 92551 29531
 45148 24854 85737 76529 77597 74392 84006 55166 68243 82128 13913 65443
 63970 94946 75461 80801 64779 12845 68647 27799 23786 88942 66916 06436
 87874 70120 96823 16713 83601 82938 58370 58630 10968 27395 51557 05627
 43417 03491 49825 84265 07505 84669 79733 67984 03232 97487 41280 09900
 30599 41944 34467 56898 89605 91483 32735 35535 53693 99219 62643 19254
 49421 12036 24190 61170 76788 56493 17377 08168 09236 06032 15645 86806
 73324 87210 45553 97082 69052 00028 07100 12441 96419 44485 00287 59439
 28867 20581 98555 38076

* This table of 100 random numbers was produced according to the following specifications: Numbers were randomly selected from within the range of 0 to 99999. Duplicate numbers were allowed.